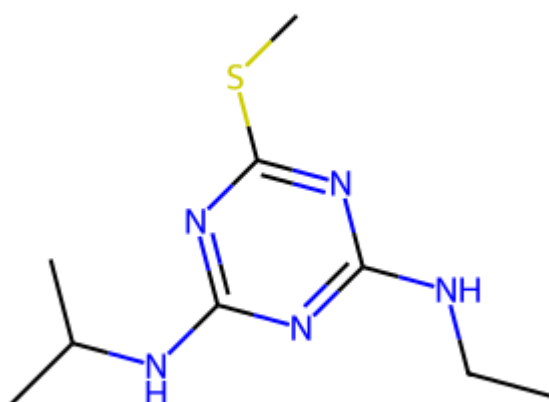


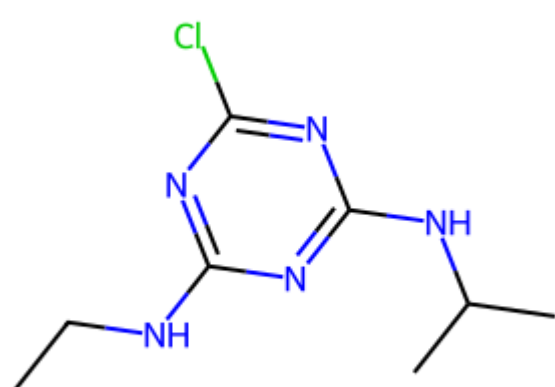


```
{'analysis_time': 0.0,  
'first_solution_time': 3.298358,  
'num_calls_rxn_model': 0,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 11.86960040890896,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 11.86960040890896,  
'retro_star_value': 11.86960040890896}
```



Reaction

```
{'analysis_time': 0.197975,  
'first_solution_time': 3.298358,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'retro_star_min_cost': 11.86960040890896,  
'retro_star_reaction_number': 11.86960040890896,  
'retro_star_rxn_cost': 6.907755278982137,  
'retro_star_value': 11.86960040890896}
```



```
{'analysis_time': 0.197975,  
'first_solution_time': 0.197975,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 11.86960040890896}
```

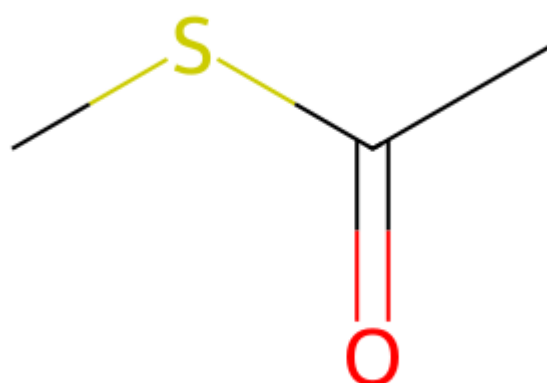


```
{'analysis_time': 0.197975,  
'first_solution_time': 3.298358,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 4.961845129926823,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 4.961845129926823,  
'retro_star_value': 11.86960040890896}
```



Reaction

```
{'analysis_time': 3.298358,  
'first_solution_time': 3.298358,  
'num_calls_rxn_model': 26,  
'num_calls_value_function': 1355,  
'retro_star_min_cost': 4.961845129926823,  
'retro_star_reaction_number': 4.961845129926823,  
'retro_star_rxn_cost': 4.961845129926823,  
'retro_star_value': 11.86960040890896}
```



```
{'analysis_time': 3.298358,  
'first_solution_time': 3.298358,  
'num_calls_rxn_model': 26,  
'num_calls_value_function': 1355,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 11.86960040890896}
```